

WEEK 3: SELLER AND INVENTORY MANAGEMENT SYSTEM

Scenario

The e-commerce company sells products supplied by multiple sellers. The system maintains seller details, tracks the products supplied by each seller, and monitors inventory levels to ensure product availability.

1. Create Seller and Inventory Tables

The **Seller** table is created to store the details of sellers, including Seller ID, Seller Name, Email, Phone Number, and Address.

The **Inventory** table is created to maintain the inventory details of products, including Inventory ID, Product ID, Seller ID, Stock Quantity, Stock Status, and Last Updated date.

```
SQL> CREATE TABLE Seller(  
2     Seller_ID VARCHAR(7),  
3     Seller_Name VARCHAR(25),  
4     Email VARCHAR(30) NOT NULL,  
5     Phone NUMBER(10),  
6     Address VARCHAR(45),  
7     CONSTRAINT PK_Seller PRIMARY KEY(Seller_ID),  
8     CONSTRAINT Un_mail UNIQUE(Email)  
9 );
```

Table created.

```
SQL> CREATE TABLE Inventory(  
2     Inventory_ID VARCHAR(7) PRIMARY KEY,  
3     Product_ID VARCHAR(7),  
4     Seller_ID VARCHAR(7),  
5     Stock_Quantity NUMBER(11),  
6     Stock_Status VARCHAR(11),  
7     Last_Updated DATE,  
8     CONSTRAINT FK_Prod  
9         FOREIGN KEY(Product_ID)  
10        REFERENCES Product(Product_ID),  
11     CONSTRAINT FK_Seller  
12        FOREIGN KEY(Seller_ID)  
13        REFERENCES Seller(Seller_ID),  
14     CONSTRAINT CK_Quan  
15        CHECK(Stock_Quantity >= 0)  
16 );
```

Table created.

2. Establish Relationships

Seller and Product

One seller can supply multiple products, and a product can be supplied by one or multiple sellers.

Relationship:

Seller → Product is **One-to-Many**.

Product and Inventory

Each product has inventory information, and the inventory table is used to track the available stock of each product.

Relationship:

Product → Inventory is **One-to-One**.

3. Constraints

Primary keys are used to uniquely identify the records in the tables.

The **Seller ID** is used as the primary key for the Seller table, while the **Inventory ID** is used as the primary key for the Inventory table.

The Inventory table also contains foreign keys to establish relationships with the Product and Seller tables.

A unique constraint is applied to the **Email** field to ensure that each seller has a unique email ID.

A check constraint is applied to **Stock Quantity** to ensure that the quantity does not contain a negative value.

4. Insert Values into Seller Table

Five seller records are inserted into the Seller table. The records contain the seller ID, seller name, email, phone number, and address.

The inserted sellers are **Lira, MJ, Blissy, Boran, and Sangyan**.

All the records are successfully inserted into the Seller table.

```
SQL> INSERT INTO Seller VALUES
  2 ('S00001','Lira','lira@gmail.com',9876543210,'Anna Nagar, Chennai');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 ('S00002','MJ','mj@gmail.com',9876543211,'T Nagar, Chennai');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 ('S00003','Blissy','blissy@gmail.com',9876543212,'Velachery, Chennai');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 ('S00004','Boran','boran@gmail.com',9876543213,'Adyar, Chennai');

1 row created.

SQL>
SQL> INSERT INTO Seller VALUES
  2 ('S00005','Sangyan','sangyan@gmail.com',9876543214,'Tambaram, Chennai');

1 row created.
```

```
SQL> SELECT * FROM Seller;
```

SELLER_	SELLER_NAME	EMAIL	PHONE
ADDRESS			
S00001	Lira Anna Nagar, Chennai	lira@gmail.com	9876543210
S00002	MJ T Nagar, Chennai	mj@gmail.com	9876543211
S00003	Blissy Velachery, Chennai	blissy@gmail.com	9876543212
S00004	Boran Adyar, Chennai	boran@gmail.com	9876543213
S00005	Sangyan Tambaram, Chennai	sangyan@gmail.com	9876543214

5. Insert Values into Inventory Table

Inventory records are inserted to maintain the stock information of different products supplied by the sellers.

The inventory records contain the **Inventory ID, Product ID, Seller ID, Stock Quantity, Stock Status, and Last Updated date.**

The records show both available and unavailable stock for different products.

```
SQL> INSERT INTO Inventory VALUES
  2  ('Inv0001', 'P2001', 'S00001', 4, 'Available',
  3  TO_DATE('2026-09-12', 'YYYY-MM-DD'));

1 row created.
```

```
SQL> SELECT * FROM Inventory;

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
-----
Inv0001 P2001 S00001 4 Available 12-SEP-26
Inv0002 P2002 S00002 3 Available 13-AUG-26
Inv0003 P2003 S00003 4 Unavailable 15-SEP-26
Inv0004 P2001 S00004 3 Available 14-SEP-26
Inv0005 P2002 S00005 5 Available 16-SEP-26
```

6. Report 1: Seller-wise Product Report

The Seller-wise Product Report displays the details of products supplied by each seller.

The report contains:

- Seller Name
- Number of Products Supplied
- Available Stock

The report uses the Seller and Inventory tables to obtain the required information.

From the result, each listed seller has supplied one product, while the available stock varies according to the inventory records.

```
SQL> SELECT s.Seller_Name,  
2         COUNT(i.Product_ID) AS Number_of_Products,  
3         SUM(i.Stock_Quantity) AS Available_Stock  
4 FROM Seller s  
5 JOIN Inventory i  
6 ON s.Seller_ID = i.Seller_ID  
7 WHERE i.Stock_Status = 'Available'  
8 GROUP BY s.Seller_Name;
```

SELLER_NAME	NUMBER_OF_PRODUCTS	AVAILABLE_STOCK
Lira	1	4
MJ	1	3
Boran	1	3
Sangyan	1	5

7. Report 2: Stock Availability Report

The Stock Availability Report provides information about the current stock of products.

It displays:

- Product Name
- Stock Quantity
- Stock Status

The report helps to identify whether the products are currently available or unavailable and shows their corresponding stock quantities.

```
SQL> SELECT p.Product_Name,
2         i.Stock_Quantity,
3         i.Stock_Status
4 FROM Product p
5 JOIN Inventory i
6 ON p.Product_ID = i.Product_ID;
```

PRODUCT_NAME

STOCK_QUANTITY STOCK_STATU

Smartphone

4 Available

Running Shoes

3 Available

Sofa

4 Unavailable

PRODUCT_NAME

STOCK_QUANTITY STOCK_STATU

Smartphone

3 Available

Running Shoes

5 Available

8. Report 3: Inventory Analysis

The Inventory Analysis report is used to analyse the overall stock maintained in the inventory.

The analysis includes:

Total Products Available

The total number of inventory records with **Available** status is **4**.

Products Out of Stock

The number of products with **Out of Stock** status is **0**.

Highest Stocked Product

The product with the highest stock quantity is **Running Shoes**, with a stock quantity of **5**.

Average Inventory Quantity

The average inventory quantity is **3.8**.

```
SQL> SELECT COUNT(*) AS Total_Available_Products
2  FROM Inventory
3  WHERE Stock_Status = 'Available';
```

```
TOTAL_AVAILABLE_PRODUCTS
-----
                        4
```

```
SQL> SELECT COUNT(*) AS OutOfStock_Products
2  FROM Inventory
3  WHERE Stock_Status = 'Out of Stock';
```

```
OUTOFSTOCK_PRODUCTS
-----
                        0
```

```
SQL> SELECT p.Product_Name,
2         i.Stock_Quantity
3  FROM Product p
4  JOIN Inventory i
5  ON p.Product_ID = i.Product_ID
6  WHERE i.Stock_Quantity = (
7         SELECT MAX(Stock_Quantity)
8         FROM Inventory
9  );
```

```
PRODUCT_NAME
```

```
STOCK_QUANTITY
```

```
Running Shoes
5
```

```
SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory
2  FROM Inventory;
```

```
AVERAGE_INVENTORY
-----
3.8
```

Conclusion

The Seller and Inventory Management System successfully maintains seller details and inventory information. The reports help to identify seller-wise products, monitor stock availability, and analyse inventory quantities. This system provides useful information for effective inventory management.